

Pei Zhang

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🕒 Last updated: May 21, 2026

Education

- 2019/09 - 2025/05  **Ph.D. in Mathematical Statistics, University of Maryland, College Park (UMD)**
Dissertation title: *Mixed Modeling Approaches for Characterizing Genetic Effects and Heritability Metrics in Longitudinal Phenotypes* (NIH Outstanding Graduate Research)
Primary Advisors: Drs. Paul S. Albert (NIH Biostatistics) and Doron Levy (UMD Applied Mathematics)
Co-advisors: Drs. Jianxin Shi (NIH Biostatistics) and Hyokyoung G. Hong (NIH Biostatistics)
- 2014/09 - 2016/06  **M.S. in Mathematics, University of Science and Technology of China (USTC)**
Dissertation title: *Ekeland Index Theory for Periodic Solutions and Symmetric Periodic Solutions of Convex Hamiltonian Systems*
Advisor: Dr. Xinan Ma
Co-advisor: Dr. Hui Liu
- 2010/09 - 2014/06  **B.S. in Mathematics and Applied Mathematics (Teacher Education Program), Nanjing Normal University (NNU)**
Dissertation title: *Some Problems in Hamiltonian Systems* (NNU Outstanding Graduation Thesis)
Advisor: Dr. Yujun Dong
- 2009/09 - 2010/06  **Undergraduate in Computer Science, NNU**

Employment History

- 2025/05 - present  **Postdoctoral Fellow (Mentored by Drs. Paul S. Albert and Jianxin Shi)**
*Biostatistics Branch (BB), Division of Cancer Epidemiology & Genetics (DCEG)
National Cancer Institute (NCI), National Institutes of Health (NIH)*
- 2021/09 - 2025/05  **Predoctoral Fellow (Mentored by Drs. Paul S. Albert and Jianxin Shi, and Hyokyoung G. Hong)**
*BB, DCEG
NCI, NIH*
- 2019/08 - 2021/08  **Teaching Assistant**
Department of Mathematics, UMD
- 2016/08 - 2017/04  **Mathematics Teacher**
NO.1 Middle School Affiliated to Central China Normal University, Wuhan, China

Employment History (continued)

2014/09 - 2016/06

■ **Teaching Assistant**
Department of Mathematics, USTC

Research Interests

- **Longitudinal and correlated data methodology**, with emphasis on repeated-measures analysis, biomarker studies, and complex dependent data structures.
- **Statistical and computational methods in genetics and genomics** for analyzing electronic health records (EHR), real-world data (RWD), and large-scale population studies (e.g., the Prostate, Lung, Colorectal, and Ovarian (PLCO) Cancer Screening Trial), with applications to prostate cancer risk stratification, early detection, and prediction.
- **Statistical and machine learning approaches in epidemiology**, including machine learning and deep learning for cancer genetic epidemiology, nutritional epidemiology, and metabolomics.

Publications

* corresponding author † co-first / equal contribution ‡ alphabetical order

Methodological / Statistical Papers

- 1 **Zhang, Pei**, X. Wang, Shi, Jianxin^{*}, and Albert, Paul S.^{*}, “Estimating the heritability of longitudinal rate-of-change: Genetic insights into PSA velocity in prostate cancer-free individuals,” *Biostatistics*, 2026, *Biostatistics* (In Press), preprint on arXiv [[arXiv](#)].
- 2 **Zhang, Pei**, Albert, Paul S.^{*}, and Hong, Hyokyoung G.^{*}, “Mixed modeling approach for characterizing the genetic effects in a longitudinal phenotype,” *Annals of Applied Statistics*, vol. 19, no. 3, pp. 2070–2087, 2025, [[PDF](#)].

Collaborative / Applied Papers

- 1 H. Fan, H. Zhao, **Zhang, Pei**, P. Yu, Y. Ji, G. Chen, H. Jin, Y. Liu, J. Liu, Z.-S. Chen, Lyu, Aiping^{*}, Liang, Xinmiao^{*}, and Chen, Yang^{*}, “Homoisoflavanone delays colorectal cancer progression via DNA damage-induced mitochondrial apoptosis and parthanatos-like cell death,” *Advanced Science*, e11406, 2026, [[PDF](#)].
- 2 H. Fan, H. Zhao, L. Gao, Y. Dong, **Zhang, Pei**, P. Yu, Y. Ji, Z.-S. Chen, X. Liang, and Chen, Yang^{*}, “CCN1 enhances tumor immunosuppression through collagen-mediated chemokine secretion in pancreatic cancer,” *Advanced Science*, e2500589, 2025, [[PDF](#)].
- 3 Loftfield, Erika^{†*}, **Zhang, Pei**[†], C. P. O’Connell, L. L. Kahle, K. Herrick, L. Abar, N. Khandpur, E. M. Steele, and H. G. Hong, “Performance of a food frequency questionnaire for estimating ultraprocessed food intake according to the NOVA classification system in the United States NIH-AARP diet and health study,” *Journal of Nutrition*, vol. 155, no. 7, pp. 2376–2384, 2025, [[PDF](#)].
- 4 Wu, Yetian[†], **Zhang, Pei**[†], H. Fan, C. Zhang, P. Yu, X. Liang, and Chen, Yang^{*}, “GPR35 acts a dual role and therapeutic target in inflammation,” *Frontiers in Immunology*, vol. 14, no. 1254446, 2023, [[PDF](#)].

Preprints / Under Review

- 1 Lim, Jungeun*, **Zhang, Pei**, J. Rhee, M. C. Playdon, A. J. Cross, R. Stolzenberg-Solomon, V. L. Roger, M. P. Purdue, Albanes, Demetrius*, Hong, Hyokyoung G.*, and Wong, Jason Y. Y.*, “Metabolomic profile score for predicting serum per- and polyfluoroalkyl substance (PFAS) levels: A novel application of machine learning to enhance detection of environmental pollutants,” Under review at *International Journal of Hygiene and Environmental Health*, 2026.

PhD Thesis

- 1 **Zhang, Pei**, “Mixed modeling approaches for characterizing genetic effects and heritability metrics in longitudinal phenotypes,” [\[PDF\]](#), Ph.D. dissertation, University of Maryland, College Park, 2025.

Honors

2026	■ National Science Foundation (NSF) Travel Support Award, USA ■ Institute of Mathematical Statistics (IMS) New Researcher Travel Award, USA
2025 & 2026	■ NIH Postdoctoral Fellowship (twice)
2025	■ NCI-DCEG Institutional Nominee for NCI Pathway to Independence Award for Outstanding Early Stage Postdoctoral Researchers (K99/R00) [about "Early K99" Grant - NIH] [about "Early K99" Grant - Stanford Medicine] [Notice of Funding Opportunity Number: PAR-23-286] ■ NIH Fellows Award for Research Excellence (FARE) 2026 (top 25% of all applicants) [List of FARE 2026 Awardees Abstracts] [about FARE] ■ NIH Outstanding Graduate Research
2023 & 2024	■ Women in Statistics and Data Science Conference Student Travel Award (twice)
2023	■ UMD Jacob K. Goldhaber Travel Grant ■ UMD International Conference Student Support Award
2022 & 2023	■ UMD Graduate Student Travel Fellowship (twice)
2022	■ Rutgers University Travel Award for Conference on Advances in Bayesian and Frequentist Statistics
2021 - 2024	■ UMD Tuition Award for External Fellowship (4 times) ■ NIH Predoctoral Fellowship (4 times) [about NIH Graduate Partnerships Program (GPP)]
2019 & 2020	■ UMD Dean’s Fellowship (twice)
2017	■ First Award in Thesis Contest on Deep Integration of Key Competence and Classroom Teaching (NO.1 Middle School Affiliated to Central China Normal University)
2016	■ USTC Outstanding Graduate
2014 & 2015	■ USTC First Award of Academic Scholarship (twice)
2014	■ NNU Outstanding Graduate ■ NNU First Award of Feng Ruer Scholarship ■ NNU Outstanding Dissertation
2013	■ Certificate of Recommendation for Exemption from Entrance Examination—USTC Graduate Program ■ Eligibility for Recommended Admission (top 5% in Mathematics Teacher Education Program, NNU) ■ Second Award of Jiangsu Teaching Professional Skills Contest (ranked 4th in Mathematics across the Jiangsu Province)
2012	■ NNU Second Award in Mathematical Modeling Contest

Honors (continued)

- 2010 & 2011
 - NNU Outstanding Debate Team Award, Debate Competition on “Is Beauty Subjective or Objective?” Member of the award-winning team arguing that beauty is objective.
- 2009
 - NNU First Award of Rolling Scholarship (twice)
 - NNU First Award in Computer Contest

Academic Activities

Invited Presentations

- 2026/07
 - International Biometric Conference (IBC), Seoul, South Korea (Invited Talk)
- 2026/05
 - University of California, Davis, Davis, California, USA (Invited Talk)
 - STATGEN 2026: Conference on Statistics in Genomics and Genetics, Emory University, Atlanta, Georgia, USA (Invited Poster)
 - The 10th Annual NIH Vivian Pinn Symposium, Bethesda, Maryland, USA (Invited Poster)
- 2026/01
 - Biostatistics Branch of NCI-DCEG Works in Progress, Rockville, Maryland, USA (Invited Talk)
- 2024/10
 - Women in Statistics and Data Science (WSDS), Reston, Virginia, USA (Invited Panelist)

Contributed Presentations

- 2026
 - The American Society of Human Genetics (ASHG) Annual Meeting, Montreal, Canada
 - Joint Statistical Meetings (JSM), Boston, Massachusetts, USA
 - The 11th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, Georgia, USA (Contributed Poster as a recipient of the IMS New Researchers Travel Award and NSF Travel Support Award)
 - AI Day for Federal Statistics, National Academy of Sciences, Washington DC, USA
- 2025
 - Joint Statistical Meetings (JSM), Nashville, Tennessee, USA
 - The 17th Annual NCI-DCEG Fellows’ Training Symposium, Rockville, Maryland, USA
- 2024
 - Joint Statistical Meetings (JSM), Portland, Oregon, USA
 - AI Day for Federal Statistics, National Academy of Sciences, Washington DC, USA
 - Probability and Statistics Day, University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA
 - NIH Annual Graduate Student Research Symposium, Bethesda, Maryland, USA
- 2023
 - Women in Statistics and Data Science (WSDS), Seattle, Washington, USA
 - Joint Statistical Meetings (JSM), Toronto, Canada
 - Probability and Statistics Day, University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA
- 2022
 - Conference on Advances in Bayesian and Frequentist Statistics with a Celebration of the 80th Birthday of Professor William E. Strawderman, Rutgers University, New Brunswick, New Jersey, USA
 - Annual Conference on Quantitative Approaches in Biology, Northwestern University, Chicago, Ohio, USA

Academic Activities (continued)

Attendant

- 2026  Artificial Intelligence (AI) Expo, The Department of Health and Human Services (HHS), Washington D.C., USA
-  NIH Library Artificial Intelligence (AI) Trainings led by OpenAI experts, Bethesda, Maryland, USA
-  Biostatistics Symposium of South California (BSSC), Newport Beach, California, USA
- 2025  Eastern North American Region (ENAR), New Orleans, Louisiana, USA
- 2024  Statistics in the Age of AI Conference, The George Washington University (GWU), Washington DC, USA
-  Eastern North American Region (ENAR), Baltimore, Maryland, USA
- 2023  National Heart, Lung, and Blood Institute (NHLBI) Biostatistics Workshop on Clinical Trial Designs: Innovative Endpoints and Futility Monitoring, Bethesda, Maryland, USA
- 2022  International Small Area Estimation (SAE) Annual Conference, UMD, College Park, Maryland, USA
- 2016  National Tianyuan Foundation Summer School on Pure Mathematics, Summer Seminar on Pure Mathematics, Zhejiang University, Hangzhou, China
- 2013  Yangtze River Delta Symposium on Partial Differential Equations and Doctoral Student Forum, Tongji University, Shanghai, China

Grant Writing Workshop

- 2026  NCI K Grant Writing Workshop, Bethesda, Maryland, USA
-  NCI-DCEG Writing Your Specific Aims Session, Rockville, Maryland, USA
- 2025  NCI-DCEG Grant Writing Workshop, Rockville, Maryland, USA

Software

R Packages and Research Code

-  **Heritability-Velocity** — Author and Maintainer. Reproducible framework for joint estimation of static and dynamic heritability components in longitudinal phenotypes, featuring scalable AI-REML and REHE algorithms for genome-wide data. [\[GitHub\]](#) | [\[arXiv\]](#)
-  **PLCO** — Author and Maintainer. Mixed-effects modeling framework with crossed genetic and individual-specific random effects to quantify genetic influences on PSA trajectories in large-scale prostate cancer screening studies. [\[GitHub\]](#) | [\[Paper\]](#)
-  **AARP** — Author and Maintainer. Implementation of measurement error models for bias correction in food frequency questionnaire-based dietary intake assessment. [\[GitHub\]](#) | [\[Paper\]](#)

Programming and Statistical Computing

-  R | Python | C/C++ | SQL | Jupyter Notebook

Research Infrastructure

-  NIH Biowulf High-Performance Computing Cluster | All of Us Research Hub | UK Biobank

Software (continued)

Scientific Writing and Reproducibility

📖 $\text{\LaTeX}$ | Microsoft Word | R Markdown | Git

Teaching

Teaching Assistant / Grader

Fall 2019	📖 Sampling Theory (STAT440) , Undergraduate/master-level. <i>Grader, UMD</i>
Spring 2020	📖 Introduction to Linear Algebra (MATH240) , Undergraduate-level. <i>Teaching Assistant, UMD</i>
Summer 2020	📖 Introduction to Mathematical Proof (MATH310) , Undergraduate-level. <i>Grader, UMD</i>
	📖 Introduction to Probability Theory (STAT410) , Undergraduate/master-level. <i>Grader, UMD</i>
Fall 2020 & Spring 2021	📖 Elementary Probability and Statistics (STAT100) , Undergraduate-level. <i>Teaching Assistant, UMD</i>
Spring 2015 & 2016	📖 Calculus II , Undergraduate-level. <i>Teaching Assistant, USTC</i>
Fall 2016	📖 Calculus I , Undergraduate-level. <i>Teaching Assistant, USTC</i>

Mathematics Teacher (Internship)

2013/09 - 2013/11	📖 High School Mathematics . High school level. <i>Mathematics Teacher Intern, Jinling High School, Nanjing, China</i>
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Service

Journal Reviewer

2021 - present	📖 Biometrics (1); Statistics in Medicine (2); Journal of Biopharmaceutical Statistics (1); Frontiers in Oncology (1); LabMed Discovery (1)
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Judging

2026	📖 Chief Judge, NIH Fellows Award for Research Excellence (FARE) 2027 Competition
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Conference / Seminar Organizer

2024	📖 Co-organizer of the panel "Discussion of Mentoring Program and Participant Experiences" at WSDS 2024
	📖 Chair of the session "Rethinking Current Modeling Strategies: Advances and Applications" at JSM 2024
2023	📖 Chair of the session "Concurrent - Statistical Insights into Mortality and Inference: From Choquet Capacities to Demographic Patterns" at WSDS 2023

Service (continued)

Mentoring

- 2025–present  Co-mentor (with **Paul S. Albert, PhD**) of postbaccalaureate fellow **Sumaja Bandreddi** (M.S., B.A., University of Virginia), Biostatistics Branch, DCEG, NCI, NIH
- Spring 2021  Directed Reading Program mentor for undergraduate **Andrew Craun**, Department of Mathematics, University of Maryland, College Park

Leadership

- 2026  Member, NIH Fellows Award for Research Excellence (FARE) 2027 Committee
- 2022–2023  Vice President, Women in Mathematics, University of Maryland, College Park
- 2020–2021  Representative, Graduate Student Government, University of Maryland, College Park
- 2010–2011  Team Leader, Volunteer Interpreters Group, Nanjing Yunjin Museum, Nanjing, China

Volunteer Service

- 2026  Volunteer, The Chinese Biopharmaceutical Association-USA (CBA)
- 2025  Volunteer, Sino-American Pharmaceutical Professionals Association (SAPA)-DC
- 2022  Volunteer Interviewee, The Center for Mathematics Education, University of Maryland, College Park
- 2009–2010  Volunteer Teacher, Nanjing Hongshan Primary School for Children of Migrant Workers, Nanjing, China

Memberships

- 2026 - present  American Society of Human Genetics
- 2024 - present  Washington Statistical Society
-  International Chinese Statistical Association
-  Section of Statistics in Genetics and Genomics
- 2023 - present  American Statistical Association
-  Institute of Mathematical Statistics
-  Society for Mathematical Biology
-  Caucus for Women in Statistics and Data Science
-  Eastern North America Region/International Biometric Society
- 2019 - 2025  American Mathematical Society

References

Paul S. Albert

Senior Investigator

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Jianxin Shi

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National Cancer Institute

Rockville, MD 20850, USA

jianxin.shi@nih.gov

Doron Levy

Professor of Mathematics

Chair, Department of Mathematics

Director, Brin Mathematics Research Center

Co-director, the NCI-UMD Partnership for Integrative Cancer Research

Fellow of the Society for Industrial and Applied Mathematics (SIAM)

University of Maryland, College Park

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